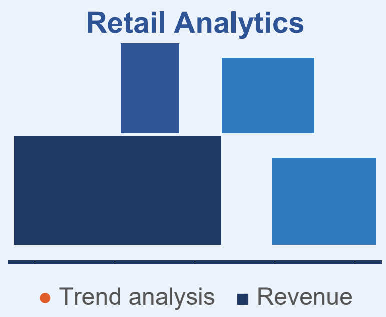
TECHNICAL REPORT

International Retail Performance Analysis

Retail customer and marketing analysis using Excel, SQL, and Tableau to identify purchasing behaviour patterns, product performance, and advertising channel effectiveness across a global business operating in eight countries.



Andrew Willacy | June 2025

TOOLS & TECH STACK

Excel
Data cleaning & EDA

SQL
Database & querying

Tableau
Dashboards & viz

KEY BUSINESS QUESTIONS

Who are the customers?
Demographics across 8 countries

What sells best?
Product category performance

Which channels work?
Advertising effectiveness by segment

CUSTOMER BEHAVIOUR, ENGAGEMENT & MARKETING PERFORMANCE ANALYSIS

Technical Report

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EXECUTIVE SUMMARY

This report presents the findings of an end-to-end customer and marketing analysis conducted for a multinational retail business operating across eight countries. Using Excel for data preparation, PostgreSQL for structured querying, and Tableau for stakeholder-facing visualisation, the analysis addressed three core business questions: who are the customers, which products generate the most revenue, and which advertising channels are most effective.

Key findings:

- Alcohol accounts for 50% of all product sales universally; a pattern consistent across all countries, marital statuses, and household compositions
- Instagram and Facebook drive the highest average customer spend among those exposed to advertising; brochure advertising is consistently the least effective channel.
- Spain generates the highest total revenue due to customer volume; Germany produces the highest average spend per customer; a commercially important distinction.
- Customer demographics are broadly consistent across markets: primarily aged 28–84 (average 55), graduate-educated, married or partnered.

The primary recommendation is to concentrate marketing investment in social media channels — particularly Instagram and Twitter (X) - while deprioritising brochure advertising, and to focus commercial energy on the alcohol and meat categories that drive the majority of revenue.

BUSINESS PROBLEM

The client wanted to improve its understanding of customer behaviour to support more effective commercial decision-making. Specifically, the analysis was commissioned to answer four questions:

- What is the demographic profile of the customer base?
- Which product categories generate the most revenue, and does this vary by customer segment?
- Which advertising channels are most effective at driving customer spend?
- Are there meaningful geographic differences in customer behaviour and sales performance?

Outputs were intended to inform marketing strategy, product focus, and geographic resource allocation at executive level.

DATA & ANALYTICAL APPROACH

Datasets

Two datasets were provided for analysis:

Dataset	Key Fields	Purpose
marketing_data.csv	Demographics, product spend by category, income, household composition, campaign response, country	Customer behaviour and segmentation analysis
ad_data.csv	Binary exposure indicators per customer for five advertising channels (Bulk mail, Twitter, Instagram, Facebook, Brochure)	Marketing channel effectiveness analysis

The two datasets were joined on the 'customer ID' field in PostgreSQL, producing a third table. This enabled cross-analysis of demographics, purchasing behaviour, and advertising exposure.

Data Cleaning (Excel)

The raw CSV files were imported into Excel for cleaning and preparation before database import. Key steps included:

- Date formatting standardised to DD/MM/YYYY
- Primary key (ID) validated for uniqueness; rows with identical entities across all fields except PK removed as implausible duplicates.
- Three records with implausible customer ages (>124 years) removed; youngest retained customer was 15 at registration - flagged as a data collection quality issue.
- Income outlier of $666,666 retained for initial analysis; excluded from income-specific analysis where it would skew averages materially.
- Inconsistent marital status categories standardised: 'Single' and 'Alone' merged; three records with non-standard entries ('YOLO', 'Absurd') removed as negligible in volume.
- Income fields cleaned of formatting characters ($, commas); note that income values appear to be a mix of annual and monthly figures. This could not be resolved without additional metadata and is noted as a limitation.
- Derived columns added: customer Age (calculated from Year_Birth) and Total_Spend (sum of all product category spend)
- Product category column labels translated from coded values to descriptive names (e.g. AmtLiq → Alcohol)
- Data types validated throughout using Excel's TYPE function

Cleaned datasets were exported as CSVs for import into PostgreSQL.

Database Structure (PostgreSQL)

A PostgreSQL database was created with two tables: Marketing_Data and ad_data. These use 'customerID' as the primary key and join field. SQL querying was used to answer the four business questions, with the two tables joined via INNER JOIN for advertising effectiveness analysis.

Notable SQL techniques applied:

- GREATEST() and LEAST() aggregate functions within CASE statements to identify highest and lowest performing product categories and advertising channels across multiple columns in a single query thus avoiding multiple subqueries.
- Boolean-to-integer casting (::INT) to enable summation of binary advertising exposure fields.

- Engineered ‘Media’ column in the ad_data table to enable average spend analysis by channel, using UPDATE statements to assign the dominant channel per customer.
- INNER JOIN implemented across tables to cross-reference customer demographics with advertising exposure and purchase behaviour.
- ROUND(AVG()) with ORDER BY for channel effectiveness ranking by average customer spend.

Dashboard Design (Tableau)

A Tableau dashboard was developed as the primary stakeholder deliverable. The design prioritised clarity and accessibility for a non-technical executive audience, using a consistent blue colour palette and intuitive chart type selection. Prototype layouts were developed on paper before build. Interactive filters allow stakeholders to explore the data without requiring analytical expertise.

Dashboard components:

- Customer age distribution
- Marital status and education level breakdown
- Geographic customer distribution and density (world map)
- Income distribution by customer count
- Total sales by product category
- Advertising campaign effectiveness by channel and country

KEY FINDINGS

1. Customer Demographics

The customer base spans eight countries with a relatively consistent demographic profile across markets. Customers range in age from 28 to 84 with an average of 55. The majority are married or partnered, and most are educated to graduate level or above.

Demographic	Profile
Age range	28 to 84 (average 55)
Gender	Broadly balanced
Marital status	Majority married or partnered
Education	Graduate level dominant (45%); Basic education smallest group (2%)
Income range	$1,730 to $162,397 (average $52,026, noting annual/monthly inconsistency)
Countries	Australia, Canada, Germany, India, Montenegro, South Africa, Spain, USA

No strong demographic variation was identified in purchasing behaviour between customer segments; product preferences and channel effectiveness patterns remained broadly consistent across age groups, education levels, and household compositions.

2. Product Sales Are Highly Concentrated

Alcohol products account for approximately 50% of all product revenue - a finding that is consistent across every country, marital status, and household composition in the dataset. This concentration is commercially significant: the top two categories (alcohol and meat) account for the substantial majority of total revenue, while vegetables and chocolate consistently underperform.

Product Category	Performance	Consistency Across Segments
Alcohol	~50% of total revenue	Universal — highest in every country and demographic
Meat	Second highest	Consistent across all segments
Fish	Mid-range	Broadly consistent
Chocolate	Low	Consistently underperforming
Vegetables	Lowest	Worst performer universally
Commodities	Low-mid	Broadly consistent

The absence of meaningful demographic variation in product preferences is itself a finding: targeted product marketing by segment is unlikely to yield significant returns. Commercial effort is better directed at channel optimisation than product-level segmentation.

3. Social Media Significantly Outperforms Traditional Advertising

Advertising channel effectiveness was assessed by cross-referencing channel exposure with customer spend data. Instagram drove the highest average spend ($1,609) among exposed customers, followed by Facebook ($1,541). Brochure advertising was the least effective channel across nearly all countries and demographic segments.

Channel	Avg Spend (Exposed Customers)	Assessment
Instagram	$1,609	Highest performing — prioritise
Facebook	$1,541	Strong performer
Brochure	$1,370	Lowest by country/demographic analysis — deprioritise
Bulk Email	$848*	Effective specifically for single customers
Twitter (X)	$848	Moderate overall; strong in specific markets
No exposure	$532	Baseline

* Bulk email performs comparably to Twitter (X) for single customers - a segment-specific finding worth exploiting in targeted campaigns.

Brochure advertising is the worst-performing channel in every country except Montenegro (two customers, statistically negligible). The case for maintaining brochure spend is weak across all markets.

Methodological note: the advertising dataset assigns one channel per customer based on the last Boolean flag recorded. Customers exposed to multiple channels are not fully accounted for in this analysis. This limitation affects the precision of spend attribution but not the directional conclusions.

4. Geographic Performance: Volume vs Value

Spain generates the highest total revenue ($652,074) by a significant margin, driven by its customer volume (994 customers - the largest market). However, Germany produces the highest average spend per customer ($694.50), despite having only 103 customers.

Country	Total Revenue	Avg Spend / Customer	No. Customers
Spain	$652,074	$656.01	994
South Africa	$204,380	$674.52	303
Canada	$163,499	$670.08	244
India	$76,619	$580.45	132
Australia	$76,048	$598.80	127
Germany	$71,533	$694.50	103
USA	$70,974	$689.07	103
Montenegro	$2,149	$1,074.50	2

While at a glance Montenegro's average spend figure ($1,074.50) seems to be the highest, it is statistically meaningless at only two customers and should not be used for decision-making. It was removed from income-related analysis to avoid distortion.

The Germany finding is commercially interesting: a smaller, higher-value customer base suggests an opportunity for targeted investment to grow volume while preserving spend-per-customer characteristics.

RECOMMENDATIONS

1. Concentrate Advertising Spend on Social Media

- Increase investment in Instagram and Facebook - both channels consistently drive the highest average customer spend.
- Discontinue or significantly reduce brochure advertising - it is the worst-performing channel across virtually every market and demographic.
- Target single customers with bulk email campaigns - this channel punches above its weight for this specific segment
- Review Twitter (X) investment by market - it performs well in specific geographies but inconsistently overall.

2. Focus Commercial Activity on Core Product Categories

- Prioritise inventory, promotions, and margin management around alcohol and meat - the categories that generate the majority of revenue.
- Review the commercial viability of vegetable and chocolate categories, which consistently underperform across all segments and geographies.
- Product-level segmentation marketing is unlikely to yield significant returns given the uniformity of purchasing behaviour across demographics.

3. Geographic Strategy

- Investigate the Germany market - the highest average spend per customer suggests a high-value customer profile worth understanding and growing.
- Review Montenegro: two customers and negligible revenue. Assess whether there is a genuine market opportunity or whether resources are better deployed elsewhere.
- Spain requires volume management rather than growth investment - the customer base is already the largest; focus should be on average spend uplift rather than acquisition.

LIMITATIONS

Limitation	Impact	Mitigation Applied
Income data inconsistency (annual vs monthly figures)	Limits reliability of income-based analysis	Flagged; average noted with caveat; median used where appropriate
Single advertising channel assumption	Customers exposed to multiple channels attributed to one only, affecting spend attribution accuracy	Directional conclusions treated as reliable; precise attribution noted as limited
Small advertising sample (292 of 2,240 customers)	Limits statistical robustness of channel effectiveness findings	Findings treated as directional; further data collection recommended
Montenegro sample size (2 customers)	Average spend figure is statistically meaningless	Excluded from comparative geographic analysis
Cross-sectional data only	No time dimension: trends over time cannot be assessed	Analysis scoped to pattern identification rather than trend projection
15-year-old customer at registration	Data collection quality concern	Record retained but flagged; no action taken beyond noting

FURTHER ANALYSIS

The following extensions would materially improve the analytical value of this work:

- Customer segmentation modelling - applying K-means clustering to segment customers by income, age, and spend behaviour to enable targeted engagement strategies by segment.
- Multi-channel advertising analysis - properly account for customers exposed to more than one channel and consider A/B testing to establish causal channel effectiveness rather than correlational.
- Longitudinal analysis - adding a time dimension to track how customer demographics and purchasing behaviour shift and assess seasonality.
- Customer lifetime value (CLV) modelling - estimating the long-term revenue value of different customer segments to prioritise acquisition and retention investment.
- Income normalisation - clarifying whether income figures are annual or monthly and normalising accordingly before using income as a predictor variable in any regression or segmentation work.
- Germany deep-dive - understanding what drives the higher average spend per customer to identify whether this pattern can be replicated in other markets.

REPOSITORY STRUCTURE

Project files are available at: github.com/AndrewWillacy/CUSTOMER-BEHAVIOUR-ENGAGEMENT-ANALYSIS-

File / Folder	Contents
marketing_data_cleaned.csv	Cleaned customer marketing dataset
ad_data_cleaned.csv	Cleaned advertising channel dataset
SQL Queries/	All PostgreSQL queries with inline comments
Tableau Dashboard.twbx	Packaged Tableau workbook (requires Tableau to open)
Dashboard.md	Link to live Tableau dashboard

Technical Report.pdf	This document
metadata.txt	Field descriptions for all dataset columns

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